MEEVA – Multi-Vendor E-Commerce Marketplace Platform

Meeva is a full-stack, web-based marketplace built with Django. It enables multiple vendors to register, list products and manage orders inside a single centralised platform while customers browse, purchase and track orders. Designed as a scalable academic production system, Meeva mirrors real-world marketplaces such as Amazon and Flipkart.

1

Roles

Customers · Vendors · Administrators

2

Goal

Simulate a production-grade multi-vendor marketplace with emphasis on security, scalability and maintainability.

Problem Statement

Traditional e-commerce often uses a single-seller model, causing limited product variety, higher infrastructure costs for small businesses and weak buyer trust. Independent sellers face technical, hosting, payment and security barriers that prevent them from launching stores.

Limited Variety

Single-seller sites struggle to offer breadth of products.

High Cost & Complexity

Hosting, payments and security raise barriers for small vendors.

Lack of Trust

No central verification makes buyers reluctant to transact.

Meeva addresses these pain points by providing a shared platform with vendor onboarding and centralised management.

Project Objectives

Centralised Marketplace

Aggregate vendors and storefronts under one platform.

Vendor Onboarding

Registration, product upload and KYC verification.

Customer Experience

Simplified browsing, cart and multi-vendor checkout.

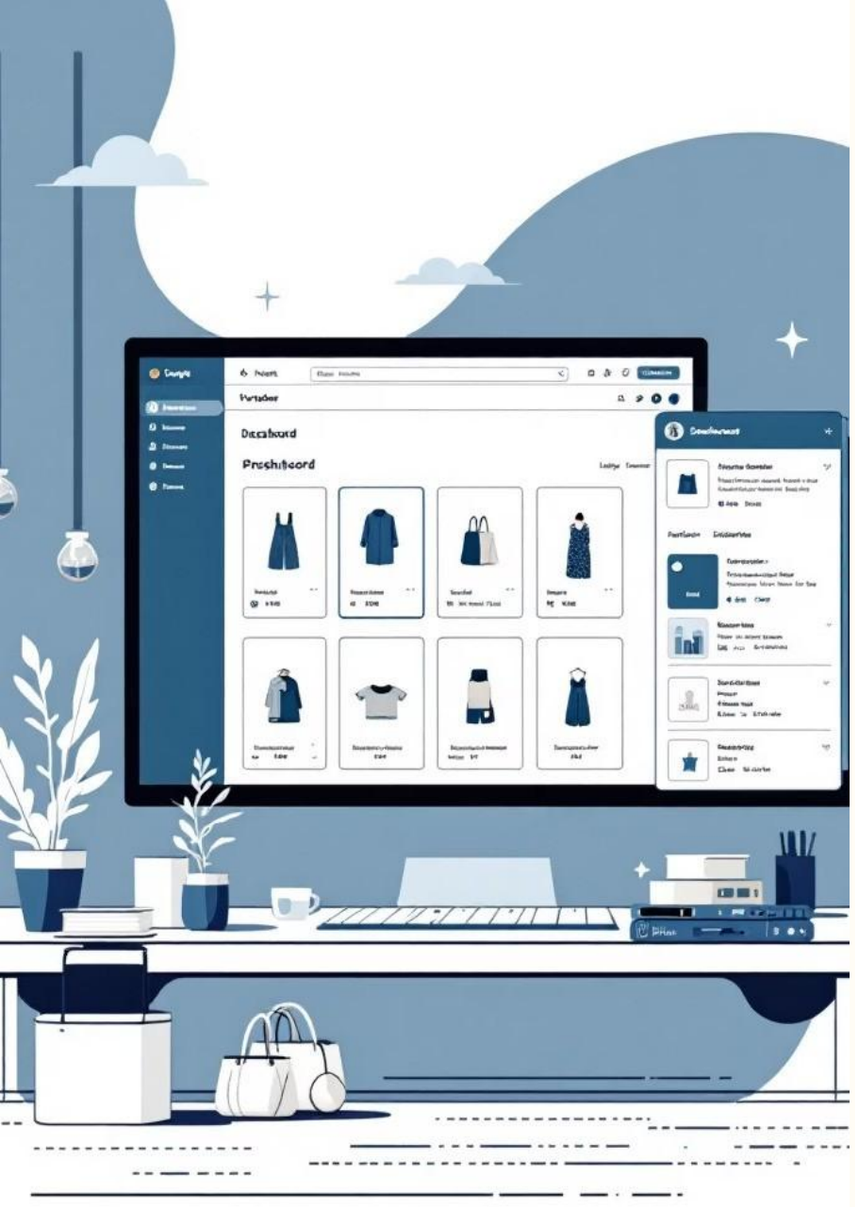
Admin Controls

Vendor approval, monitoring and notifications.

Security & Data Handling

Secure authentication, hashed passwords and validated inputs.

Secondary focus: usability, security, scalability and maintainability to support academic-level production workloads.



Scope – Included & Excluded Features

Included

- Customer registration & login
- Vendor onboarding with KYC
- Product listing & management
- Shopping cart & order placement
- Vendor dashboard
- Admin vendor approval
- Email notifications

Excluded (Future Work)

- Online payments
- Product reviews
- Mobile application
- Advanced search filters
- Real-time notifications

The current scope prioritises core marketplace flows and vendor verification; premium features are roadmap items.

Technology Stack

Django was chosen for built-in authentication, rapid development and a secure, opinionated framework with ORM abstraction.

Backend

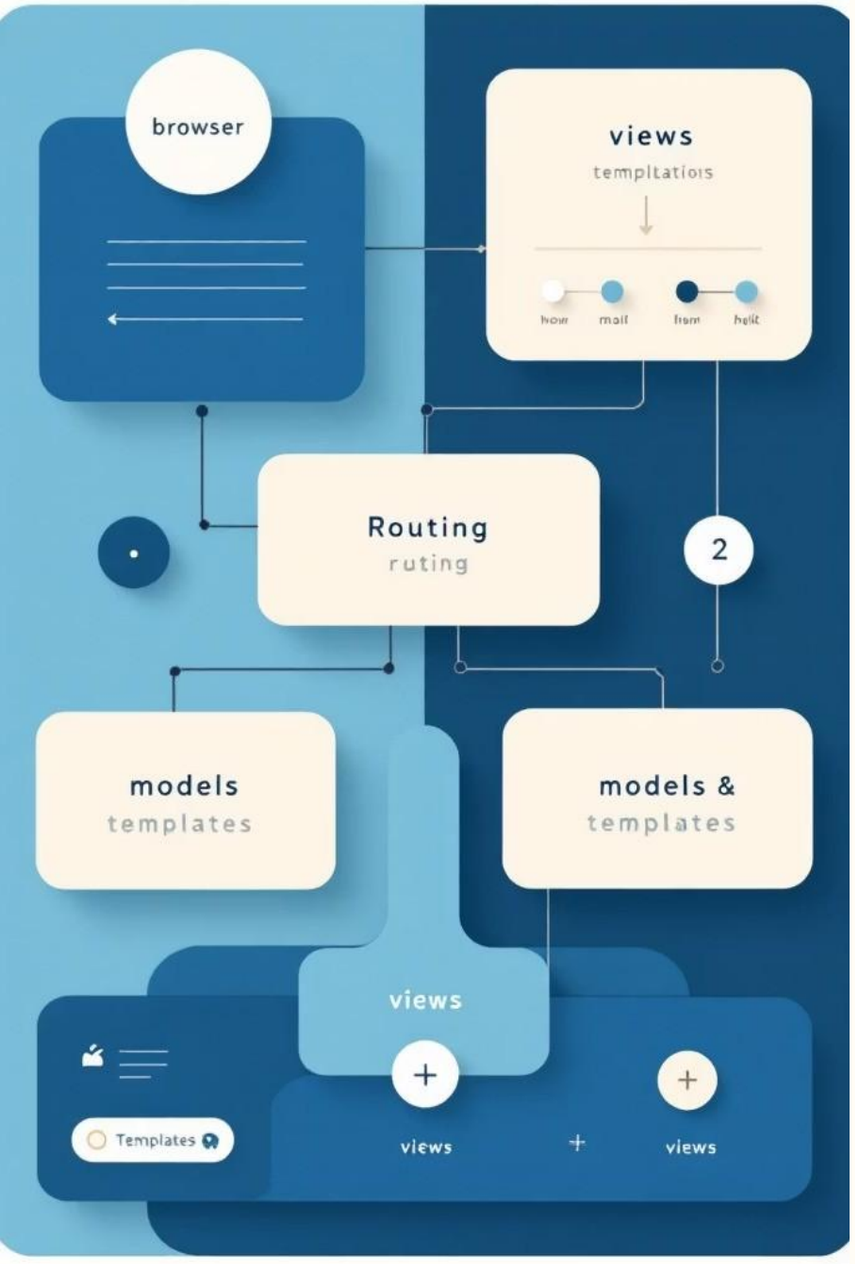
Python · Django 5.2.10 · Django ORM · SQLite

Frontend

HTML · Tailwind CSS · Font Awesome · Responsive UI

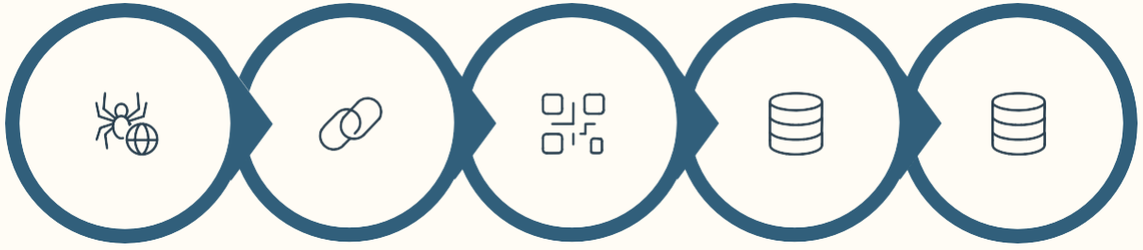
Supporting Tools

Pillow (images) · SMTP email · Environment variables (.env)



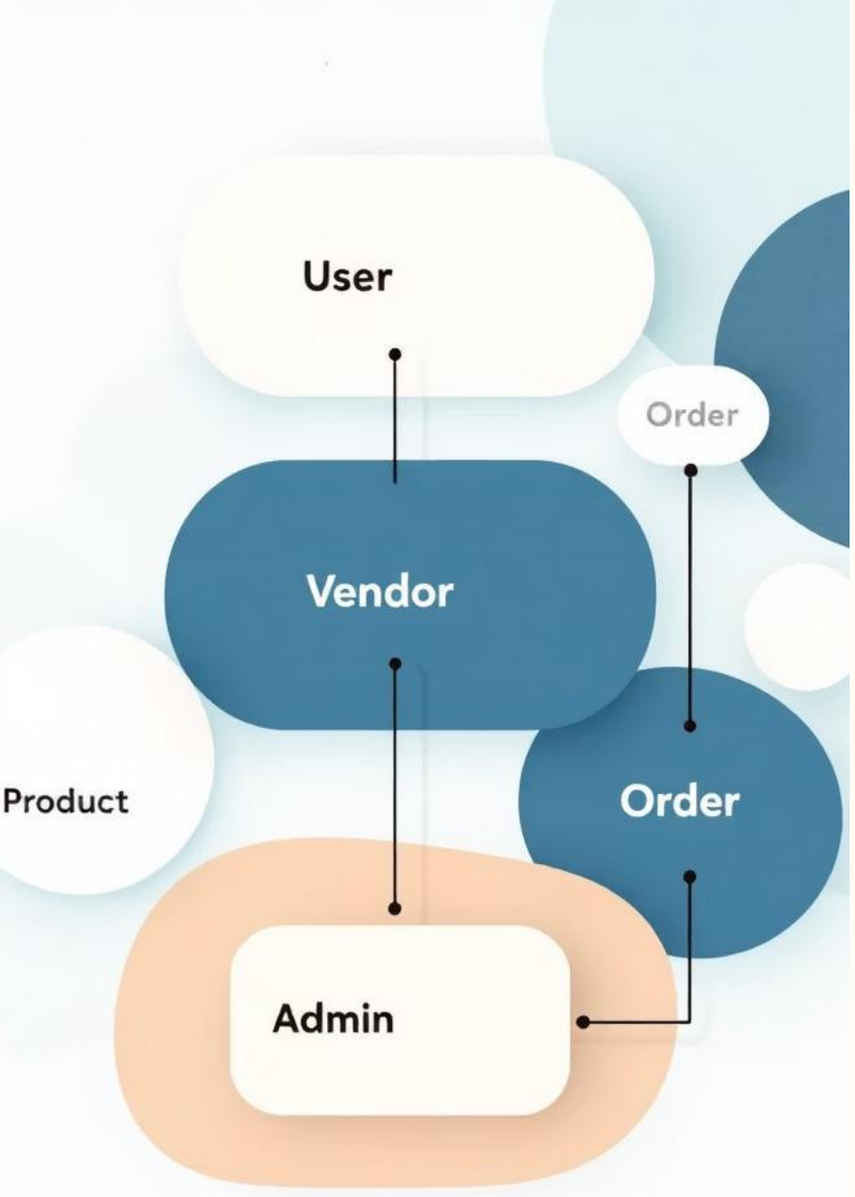
System Architecture

Meeva adheres to the Django MVT pattern. Request flow: Browser →URL routing → Views →Models →Database →Templates →Response. This clear separation enhances testability and maintainability.



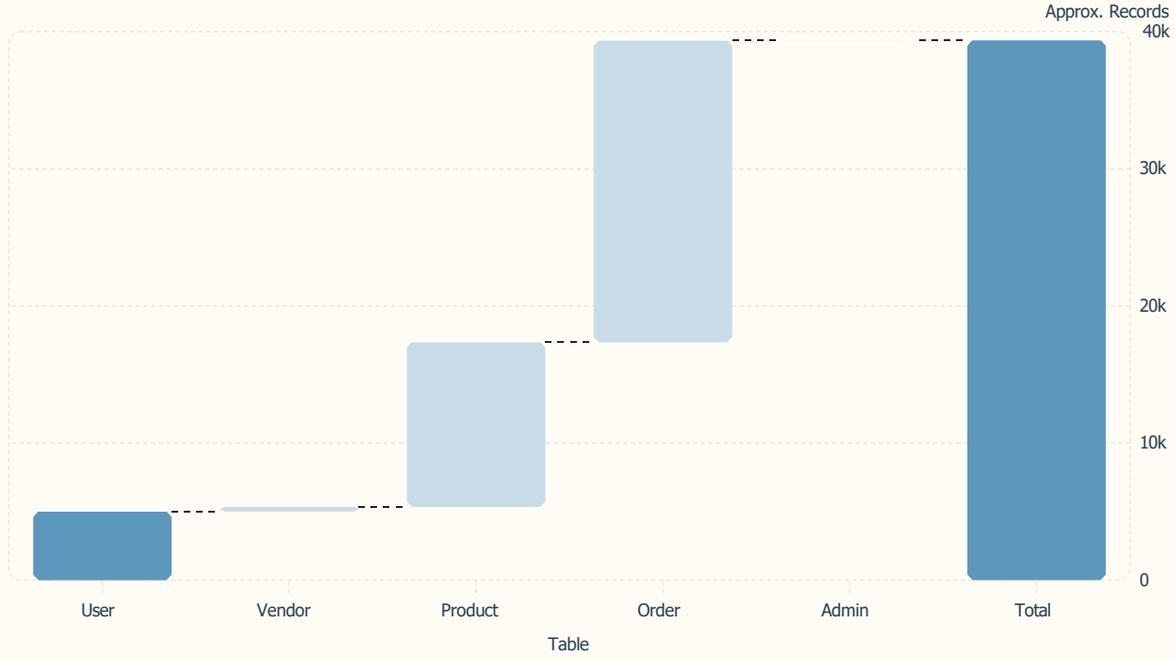
Modular Apps

Pages, Users, Vendor, Admin — each app isolates responsibility and simplifies scaling and testing.



Database Design

The schema contains five primary models: User, Vendor, Product, Order and Admin. Relationships enforce marketplace semantics and data integrity.



Key relationships: One Vendor →Many Products; One Vendor →Many Orders; One Product →Many Orders. Constraints include unique emails and validation rules to maintain data integrity.

Vendor Onboarding & KYC Verification

Step 1 – Registration

Vendor completes registration form and provides profile details.

Step 2 – Document Upload

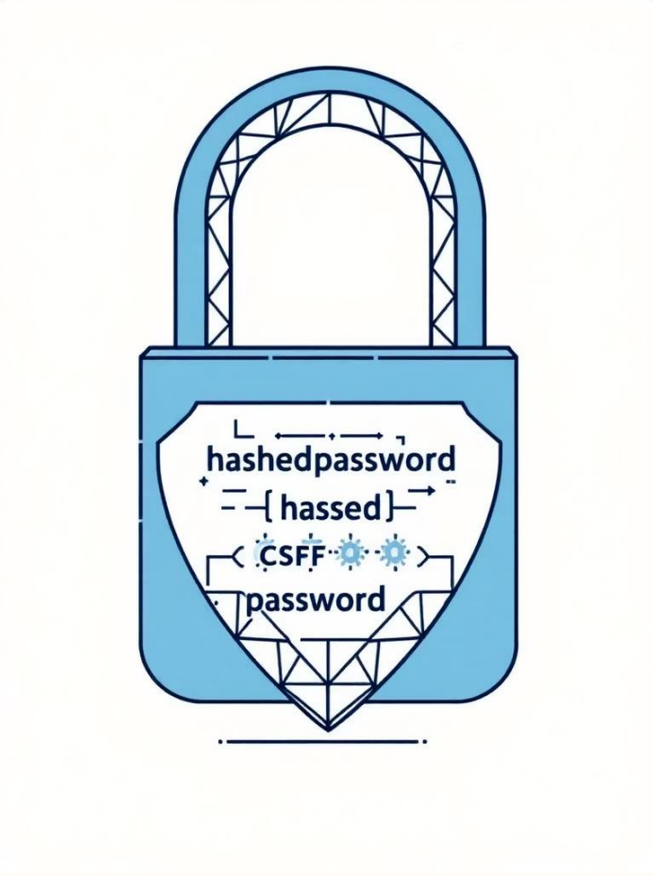
Upload Aadhar, PAN and Business License; images stored via Pillow handling.

Step 3 – Admin Review

Admin reviews documents, sets vendor status to Pending → Approved or Rejected.

This manual verification process minimises fraud, increases buyer trust and ensures only authenticated sellers list products.

Authentication & Security



- Session-based authentication with separate login flows for Customer, Vendor and Admin.
- Password hashing and CSRF protection enabled by Django defaults.
- Session management and role-based access controls enforce separation of privileges.
- Input validation and environment variables secure sensitive configuration.

Each role receives a dedicated dashboard post-login, ensuring tailored functionality and access limits.

Customer Workflow

Customers can aggregate items from multiple vendors in a single cart, complete checkout and track orders. The platform delivers a familiar, streamlined shopping path while backend services route vendor orders and notifications appropriately.

Tip: Multi-vendor cart handles per-vendor order grouping at fulfilment while keeping a unified customer checkout experience.

Vendor Dashboard & Product Management

After approval, vendors operate a full product lifecycle: add, edit and delete listings; view and manage orders; update order status; and track revenue. The interface mirrors real marketplace workflows and supports rapid product onboarding.

Add & Manage Products

Create rich product entries with images, descriptions, categories and stock levels.

Order Lifecycle

Pending → Confirmed → Shipped → Delivered — status updates drive notifications and inventory changes.

Revenue & Insights

Per-vendor revenue tracking and basic sales analytics to support business decisions.

Admin Panel — Platform Governance

The admin is the trust authority: overseeing vendor lifecycle, enforcing policies, and monitoring system health. Centralised controls ensure marketplace integrity and rapid incident response.

Review Applications

Validate vendor identity, documentation and compliance before activation.

Suspend / Reactivate

Temporarily remove malicious or non-compliant vendors, with audit trails for reinstatement.

Monitor Activity

System-wide logs and dashboards surface anomalies and enable timely interventions.

Shopping Cart & Order System

Meeva uses a session-based cart with robust checks: multi-product support, stock validation, bulk checkout and automatic order creation. Post-checkout actions ensure inventory integrity and vendor notification.

Cart Session

Session-backed cart preserves selections across browsing and short interruptions.

Validation

Real-time stock checks prevent overselling and inform users of availability changes.

Order Creation

Successful checkout triggers order creation, inventory decrement and vendor email notification.

Security Measures — Defence-in-Depth

Security is integral: layered protections and best practices reduce attack surface and protect user data. Sensitive credentials are stored in environment variables to prevent leakage.

Hashed Passwords (PBKDF2)

Adaptive hashing with salts to protect stored credentials against brute-force attacks.

CSRF & Session Protection

Anti-CSRF tokens and secure session handling prevent cross-site request forgery and session hijacking.

Input & Ownership Validation

Server-side validation ensures data integrity; ownership checks prevent unauthorised modifications.

XSS Prevention

Output encoding and strict content policies mitigate injection of malicious scripts.

Testing & Reliability

Comprehensive testing validated core flows and system stability. Automated checks and manual tests ensured readiness for further development and deployment.

Form & Authentication Tests

Validation rules, edge cases and login sequences were exercised to prevent regressions.

Cart & Order Tests

Cart persistence, stock validation and order placement were tested across scenarios.

Vendor Workflow Tests

Approval, suspension and audit trails were verified to maintain marketplace trust.

Django System Checks

All Django system checks passed with zero critical errors, confirming baseline health.

Future Enhancements — Roadmap

Planned improvements focus on payment, discovery, scalability and mobile access. These enhancements will elevate user experience and prepare the platform for production loads.

Online Payments

Integrate Razorpay / Stripe for secure, PCI-compliant payments and refunds.

Search & Filters

Advanced search, faceted filters and relevance ranking for better product discovery.

Analytics & Reviews

Vendor analytics, user reviews and ratings to drive trust and data-driven decisions.

Scalability & APIs

REST API for mobile apps, AWS deployment, Redis caching and PostgreSQL for production readiness.

Conclusion — Meeva Summary

Meeva demonstrates a production-oriented marketplace built with Django: multi-role access, vendor verification, complete shopping lifecycle, and a governance-enabled admin panel. The architecture is designed for scalability, security and extensibility—providing a solid foundation for further development.

Core Achievements

Multi-role system, vendor workflows, and full order lifecycle implemented.

Technical Strengths

Secure authentication, robust testing, and clear deployment roadmap.

Next Steps

Integrate payments, expand analytics, and deploy to cloud for production trials.